1 Core content

1.1 The impact of new and emerging technologies

1.1.1 Industry

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- Modern factories increasingly use **automated production**. Robots are often used to complete some repetitive and monotonous tasks previously carried out by humans, which can lead to unemployment.
- **Computer-aided manufacture (CAM)** machines are used where high volumes of identical products of a consistent high quality are needed.
- People often move to different countries to find suitable employment or where there is a shortage in a particular workforce. This is referred to as **demographic movement**.
- Science and technology parks are areas where like-minded businesses come together to form centres of excellence.

Automated production: a production method of using machinery controlled by computers.

CAM: computer-aided manufacture.

1.1.2 Enterprise

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- A privately-owned business is owned by the company founders, or their families, or by a small group of investors whose shares in the company are not publicly traded.
- Crowd funding is an internet-based scheme allowing people to raise money to get a business idea started, such as the manufacture of a new product. Investors lend money to projects that they believe are viable.
- The UK government offers some loans and grants to new start-up businesses. Investors can get tax relief on their investments. While not all new businesses are successful, the potential to boost the economy and the provision of future employment is worth taking a risk.
- Not-for-profit organisations reinvest all profits back into the business for the benefit of all involved.



Figure 1.1 The Forest Stewardship Council® (FSC®) is an example of a not-for-profit organisation

1.1.3 Sustainability

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- **Sustainability** is about meeting today's needs without compromising the needs of future generations.
- Demand on the world's natural resources currently exceeds supply. Materials used in product design fall into two categories:
  - **finite**: resources with a limited supply that will eventually run out
  - **non-finite**: resources that can be replenished and are unlikely to run out.
- One way to reduce our environmental impact is to consider the six Rs.

Sustainability: the manufacture of goods without compromising future needs; the materials required for a product can be replenished and will not run out.

Table 1.1 The six Rs of sustainability

Rethink	Is there a better way of making the product that is less harmful to the environment? Can the design be simplified to make manufacturing easier?
Recycle	Can the product be recycled easily after it is no longer needed? Can the components and materials be separated easily? Can recycled materials be used?
Repair	Can this product be repaired easily if it breaks? Can the component parts be replaced easily?
Refuse	Consumers might not buy a product if it is not environmentally friendly or if it is made by people working in poor conditions.
Reduce	Can the number of component parts, new materials or packaging be reduced? Can the manufacturing process be simplified to reduce the energy used?
Reuse	Can any parts be reused after it is no longer needed? Could the product be reused for something else once its primary use is no longer required?

Transportation

- Each product has a **carbon footprint** from its manufacture and the distance it travels until it is purchased and used.
- Transportation costs/carbon footprint can be reduced by:
  - sourcing materials closer to manufacturing centres and potential markets
  - using energy-efficient means of transporting goods, for example electric or **hybrid vehicles**.

Carbon footprint: a measure of the greenhouse gases emitted by human activity.

Hybrid vehicles: vehicles that are both petrol and electric, so emissions are reduced.

Pollution

The manufacture and use of any product leads to an increase in pollution levels. There are ways to reduce the level of pollution.

- Manufacturers could use cleaner energy sources, such as solar or wind.
- Goods could be transported using electric vehicles that do not emit greenhouse gases.
- Products could be built to last and easy to repair, causing less waste.
- Products need to be disposed of carefully if they cannot be recycled, avoiding landfill where possible.



Figure 1.2 Hybrid technology reduces carbon emission from vehicles

Demand on natural resources

- Natural resources and productive land enable us to produce the goods and services that sustain and support modern lifestyles.
- The **ecological footprint** is a measure of the impact that human activity has on the environment.
- Humanity's ecological footprint is currently 1.7 Earths – the Earth takes about 18 months to replace what we use in 12 months. If this continues, we will create an **ecological deficit**.

Ecological footprint: a measure of the impact that human activity has on the environment.

Ecological deficit: when resources are being used up faster than nature can replenish them.

Waste generated

- Many products are designed with built-in or planned **obsolescence**, meaning that they are not designed to be long lasting.
- Companies such as Apple introduce new products that consumers are eager to own, even though their current products still work perfectly well. This puts further strain on the world's resources, as many products are not recycled or are difficult to recycle.
- Some polymers are difficult to recycle. New technology is being developed that will allow them to be broken down more effectively and safely.

Obsolescence: the process of becoming no longer needed or wanted.

1.1.4 People

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- Developments in mobile technology and the internet make it easier for us to communicate with people all over the world. This leads to greater competition among manufacturers.
- There are downsides to this global society, which affect workers directly:
  - Importing cheap products from overseas rather than buying locally produced products can lead to job losses in our own society.
  - The use of automation leads to job losses (see Section 1.1.1 Industry on page 7).
  - Workers overseas are often paid low wages to reduce costs and maximise profit.
- The workforce of the future needs to be more adaptable, offering different skills, such as problem solving, which lead into more highly skilled jobs like engineering or computer science.
- Technological developments in **computer-aided design (CAD)** packages have changed the way designers work. All aspects of developing design ideas can be done using computers.
- Apprenticeships are work-based training programmes that allow trainees to learn on-the-job from a skilled employer. Apprenticeships are often in vocational areas, for example tailoring, engineering or carpentry.
- Society needs to be inclusive. New and emerging technologies have allowed designers to create products that meet the needs of many people, including those with disabilities and the elderly. This could be from simple hand tools to computer programs.
- Children are increasingly exposed to technology. While this can be positive, many believe it is at the expense of social interaction.

CAD: computer-aided design; software used to produce drawings and virtual models of products and systems.

1.1.5 Culture

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- People have always travelled to find better work opportunities and improve their lives. A permanent move is called **migration**. The European Union allows EU nationals to travel freely across the member states.
- Social segregation can occur when ethnic minorities cluster together to form their own society but do not fully integrate into the wider community.

Migration: the movement of people, often to find work.

Advantages and disadvantages

- The choice of materials impacts the success of the final product.
- The advantages of a particular material may make it an obvious choice for a product, but the designer must also consider disadvantages, for example a thermosetting polymer may have the best properties for a product but it is not recyclable. Therefore, the designer may choose a thermoforming polymer with properties as close to the thermosetting one as possible.

Justification of the choice

- The designer must make informed choices based on these characteristics and other factors, such as:
  - cost – is there another material available that may cost less?
  - availability – is the material in good supply?
  - sustainability – is the material sustainable?
  - environmental impact – can the material be recycled?
  - social, cultural and moral considerations – will the use of this material have a negative impact?
- When designing an electronic system, the designer must consider the range of components available and decide on the most suitable type for the application.

Now test yourself

TESTED

1 State **four** important mechanical properties of materials. (4)
2 State **two** factors that can influence and affect the properties of materials. (2)

1.14 Environmental, social and economic challenges

1.14.1 Respect for different social, ethnic and economic groups

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Positive and negative impacts of products

- Products can have a negative impact on society that the designer may not have foreseen.
- For example, leaflets and flyers can be used to communicate useful information to people about services in the area, however this can lead to unwanted junk mail being delivered, urging people to buy things they don't want.
- While it is not always easy to predict the effects of a product, designers must try to consider the possible negative implications and decide whether the product is worthwhile.

Cultural awareness

- **Cultural awareness** is considering how different cultural groups may be affected by a product. Different cultures may view products in a totally different light depending on their beliefs, ideas and experience.
- A product designed for a particular market sector should not be offensive or upsetting to another.
- Designing new products that continue traditional processes, skills and materials can help preserve cultural identity.

Cultural awareness: considering how different cultural groups may be affected by a product.

Economic challenges

- The manufacturing processes, transportation, use and disposal of a product can affect the economic needs of people.
- The manufacture of a new product may create jobs in a factory or geographical area, boosting local people's income and improving the overall economy of the area.
- If the product is manufactured with modern CAM machines, it could lead to the loss of jobs and, in turn, affect the whole economy in that area.
- Many products are made abroad because labour costs are much cheaper in developing countries.
- Some workers in these countries are poorly paid, work in dangerous or unhealthy conditions, and work long hours without breaks or holidays.
- In some countries there is use of child labour, with children working in unsafe conditions because they have no choice.
- Designers have a responsibility to ensure that people are not being exploited in this way.

1.14.2 Appreciation of the environmental, social and economic issues

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- Designers should make sure their clients follow the guidelines for the fair treatment of workers set out by organisations such as the European Institute for Crime Prevention and Control, the Gangmasters and Labour Abuse Authority (GLAA) or the Alliance for Workers Against Repression Everywhere (AWARE).
- Manufacturers of fairtrade products ensure that their workers are treated and paid fairly, and that they do not work in dangerous or unhealthy conditions (see Section 1.2.3 Ethical perspectives on page 12).
- Carbon offsetting is when companies that use high amounts of energy or produce a lot of waste 'offset' this by setting up or donating funding to projects that help to reduce greenhouse gases.
- Products that can be disassembled easily allow the components to be reused instead of throwing away perfectly useable parts and manufacturing new ones. Making it easier to separate parts for recycling means that less materials are disposed of in landfill.

1.14.3 Green design

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- Social pressure to have the latest products and the influence of fashion and trends mean that people throw away rather than repair things. This is known as the **throwaway society**.
- **Green design** means using eco-friendly materials and construction practices that safeguard air, water and earth.
- **Eco-design** means designing sustainable products that will not harm the environment by considering the effects of the technology, processes and materials used.

Throwaway society: buying products for brief use then disregarding them or throwing them away with little thought of the environmental impact.

Green design: using eco-friendly materials and practices to design products.

Eco-design: designing sustainable products that have a minimal effect on the environment.

Stakeholder: a person other than the main user who comes into contact with or has an interest in the product.

1.14.4 Recycling and reusing materials or products

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- Choose materials that are sustainable, recyclable and non-toxic, and which do not require as much energy to process.
- Design products to last as long as possible, so there is less replacement of parts, and that can be fully recycled at the end of their useful life.
- Design products that function to the best of their potential.
- Consider the needs and wants of all the **stakeholders** involved.

Typical mistake

Sustainable products and materials are not the same as recyclable products and materials.

1.14.5 Human capability

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- Human capability is the range of functions that humans can carry out.
- Cognitive capability is the ability to process thoughts, learn new things and solve problems.
- Physical capability is the ability to perform physical tasks, such as walking, standing, balancing or gripping items.
- As we mature from children into adults, our cognitive and physical capabilities develop and increase.
- Old age, illness or other events in our life can have a huge effect on our human capability.

1.14.6 Cost of materials

REVISED

- Designers must try to keep material costs as low as possible without compromising the quality of the final product.
- Where materials are expensive, designers must look for cheaper alternatives with similar properties.
- Designers must consider whether a particular material is necessary or just for 'show'.

1.14.7 Manufacturing capability

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- Manufacturing capability is what a company or manufacturing plant is able to produce.
- Many companies have approved ways of producing a particular product.
- Manufacturing capability relates to the:
  - technological capability of the manufacturer – how up to date the equipment is

- physical size and weight of the product being manufactured
- production capacity of the manufacturing plant.

1.14.8 Life-cycle analysis

REVISED

- The life cycle of a product is the stages a product goes through from beginning to end.
- A life-cycle analysis (LCA) assesses the environmental impacts of a product at the following stages:
  - extraction of the raw materials
  - processing of the materials into useable form
  - manufacture of the product
  - distribution and transportation to retailers
  - use by the customer
  - repair and maintenance of the product
  - disposal or recycling of the product.

Exam tip

Practise drawing LCAs for a range of products.

Now test yourself

TESTED

1 State **three** factors that may be considered when designing recyclable products. (3)

2 Name **three** sustainable materials and explain why they are sustainable. (6)

3 Describe **one** factor that can influence human capability. (2)

1.15 The work of past and present professionals and companies

1.15.1 Analysing a product to specification criteria

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Table 1.13 Specification criteria

Criteria	Explanation
Form	The shape and appearance of a product
Function	The purpose of a product; what it will be used for
Client/user requirements	The needs, wants and values that are important to the people who will use it
Performance requirements	Factors that will help it to do its job; how it will perform when in use
Materials and components/ systems	What the product is made from and why; the properties the materials have that support function; how the product operates (system)
Scale of production and cost	The number of products to be manufactured will affect the scale of production and costs
Sustainability	The environmental impact from origin to disposal
Aesthetics	The visual appearance of the product, for example colour, texture
Marketability	How likely the product will sell; features that will help product sales
Innovation	Does it have any radically new features or include the latest technology?